

138. Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8

138.1 Overview

This clause specifies the 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 PMDs together with the multimode fiber medium. The optical signals generated by these PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMD sublayers provide point-to-point 50, 100, 200, and 400 Gigabit Ethernet links over one, two, four, or eight pairs of multimode fiber, with a reach of up to at least 100 m.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA, as shown in Table 138–1, Table 138–2, or Table 138–3, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Table 138–1—Physical Layer clauses associated with the 50GBASE-SR PMD

Associated clause	50GBASE-SR
132—RS	Required
132—50GMII ^a	Optional
133—PCS	Required
134—RS-FEC	Required
135—PMA for 50GBASE-R	Required
135B—LAUI-2 C2C	Optional
135C—LAUI-2 C2M	Optional
135D—50GAUI-2 C2C	Optional
135E—50GAUI-2 C2M	Optional
135F—50GAUI-1 C2C	Optional
135G—50GAUI-1 C2M	Optional
78—Energy-Efficient Ethernet	Optional

^aThe 50GMII is an optional interface. However, if the 50GMII is not implemented, a conforming implementation behaves functionally as though the RS and 50GMII were present.

Table 138–2—Physical Layer clauses associated with the 100GBASE-SR2 PMD

Associated clause	100GBASE-SR2
80—RS	Required
80—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC ^b	Required
83—PMA for 100GBASE-R	Optional
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
83E—CAUI-4 C2M	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135E—100GAUI-4 C2M	Optional
135F—100GAUI-2 C2C	Optional
135G—100GAUI-2 C2M	Optional
78—Energy-Efficient Ethernet	Optional

^aThe CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^bThe option to bypass the Clause 91 RS-FEC correction function is not supported.

Table 138–3—Physical Layer clauses associated with the 200GBASE-SR4 and 400GBASE-SR8 PMDs

Associated clause	200GBASE-SR4	400GBASE-SR8
117—RS	Required	Required
117—200GMII ^a	Optional	Not applicable
117—400GMII ^a	Not applicable	Optional
118—200GMII Extender	Optional	Not applicable
118—400GMII Extender	Not applicable	Optional
119—PCS for 200GBASE-R	Required	Not applicable
119—PCS for 400GBASE-R	Not applicable	Required
120—PMA for 200GBASE-R	Required	Not applicable
120—PMA for 400GBASE-R	Not applicable	Required
120B—200GAUI-8 C2C	Optional	Not applicable
120B—400GAUI-16 C2C	Not applicable	Optional
120C—200GAUI-8 C2M	Optional	Not applicable
120C—400GAUI-16 C2M	Not applicable	Optional

Table 138–3—Physical Layer clauses associated with the 200GBASE-SR4 and 400GBASE-SR8 PMDs (*continued*)

Associated clause	200GBASE-SR4	400GBASE-SR8
120D—200GAUI-4 C2C	Optional	Not applicable
120D—400GAUI-8 C2C	Not applicable	Optional
120E—200GAUI-4 C2M	Optional	Not applicable
120E—400GAUI-8 C2M	Not applicable	Optional
78—Energy-Efficient Ethernet	Optional	Optional

^aThe 200GMII and 400GMII are optional interfaces. However, if the appropriate interface is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII or 400GMII were present.

Figure 138–1 shows the relationship of the PMDs and MDIs (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 50 Gigabit Ethernet is introduced in Clause 131 and the purpose of each PHY sublayer is summarized in 131.2. 100 Gigabit Ethernet is introduced in Clause 80 and the purpose of each PHY sublayer is summarized in 80.2. 200 Gigabit Ethernet and 400 Gigabit Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2.

50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 PHYs with the optional Energy-Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

Further relevant information may be found in Clause 1 (terminology and conventions, references, definitions and abbreviations) and Annex A (bibliography, referenced as [B1], [B2], etc.). The 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 sublayers provide point-to-point 50, 100, 200, and 400 Gigabit Ethernet links over one, two, four, or eight pairs of multimode fiber, up to at least 100 m.

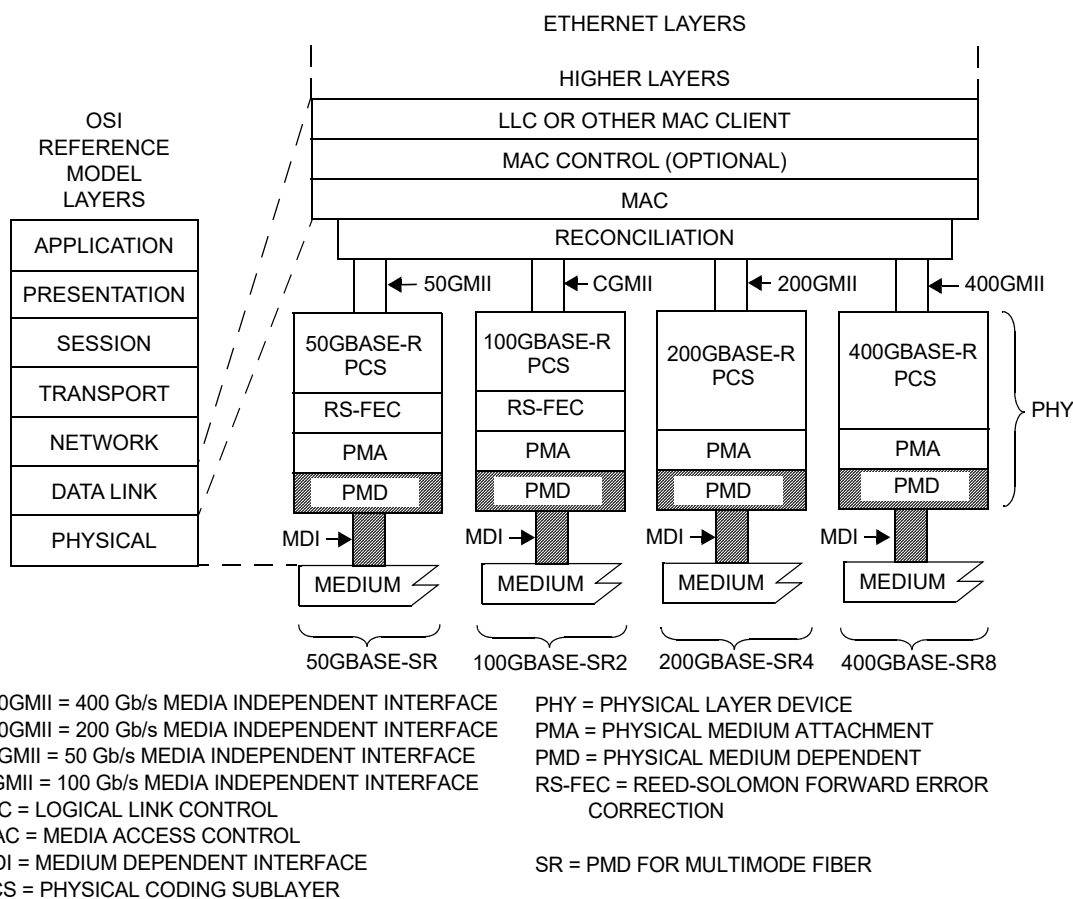


Figure 138–1—50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

138.1.1 Bit error ratio

For the 50GBASE-SR and 100GBASE-SR2 PMDs, the bit error ratio (BER) when processed by the PMA (Clause 135) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 9.2×10^{-13} for 64-octet frames with minimum interpacket gap when additionally processed by the FEC (Clause 134 or Clause 91) and PCS (Clause 133 or Clause 82). For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-10} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 9.2×10^{-13} for 64-octet frames with minimum interpacket gap.

For the 200GBASE-SR4 and 400GBASE-SR8 PMDs, the bit error ratio (BER) when processed by the PMA (Clause 120) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap when additionally processed by the PCS (Clause 119). For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this

requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.

138.2 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The 50GBASE-SR PMD service interface is an instance of the inter-sublayer service interface defined in 131.3, with a single symbol stream ($n = 1$).

The 100GBASE-SR2 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with two parallel symbol streams ($n = 2$).

The 200GBASE-SR4 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with four parallel symbol streams ($n = 4$).

The 400GBASE-SR8 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with eight parallel symbol streams ($n = 8$).

The service interface primitives are summarized as follows:

PMD:IS_UNITDATA_*i*.request
PMD:IS_UNITDATA_*i*.indication
PMD:IS_SIGNAL.indication

The 50GBASE-SR PMD has a single symbol stream, hence $i = 0$.

The 100GBASE-SR2 PMD has two parallel symbol streams, hence $i = 0$ to 1.

The 200GBASE-SR4 PMD has four parallel symbol streams, hence $i = 0$ to 3.

The 400GBASE-SR8 PMD has eight parallel symbol streams, hence $i = 0$ to 7.

In the transmit direction, the PMA continuously sends n streams of PAM4 symbols to the PMD, one per lane, using the PMD:IS_UNITDATA_*i*.request primitive, at a nominal signaling rate of 26.5625 GBd. The PMD converts these streams of symbols into appropriate signals on the MDI.

In the receive direction, the PMD continuously sends n streams of PAM4 symbols to the PMA, corresponding to the signals received from the MDI, one per lane, using the PMD:IS_UNITDATA_*i*.indication primitive, at a nominal signaling rate of 26.5625 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable SIGNAL_DETECT parameter as defined in 138.5.4. The SIGNAL_DETECT parameter can take on one of two values: OK or FAIL. When SIGNAL_DETECT = FAIL, the rx_symbol parameters are undefined.

NOTE—SIGNAL_DETECT = OK does not guarantee that the rx_symbol parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_DETECT = OK indication and still not meet the BER defined in 138.1.1.

138.3 Delay and Skew

138.3.1 Delay constraints

An upper bound to the delay through the PMA and PMD is required for predictable operation of the MAC Control PAUSE operation.

The sum of the transmit and receive delays at one end of the link contributed by the 50GBASE-SR PMD including 2 m of fiber in one direction shall be no more than 1024 bit times (2 pause_quanta or 20.48 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 100GBASE-SR2 PMD including 2 m of fiber in one direction shall be no more than 2048 bit times (4 pause_quanta or 20.48 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-SR4 PMD including 2 m of fiber in one direction shall be no more than 4096 bit times (8 pause_quanta or 20.48 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-SR8 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause_quanta or 20.48 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta, can be found in 131.4 for 50GBASE-SR, in 80.4 for 100GBASE-SR2, and in 116.4 and its references for 200GBASE-SR4 and 400GBASE-SR8.

138.3.2 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

138.3.2.1 Skew Constraints for 50GBASE-SR

Skew and Skew Variation are defined in 131.5 and specified at the points SP0 to SP7 shown in Figure 131-3.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by 135.5.3.5. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 131.5. The measurements of Skew and Skew Variation are defined in 89.7.2 with the exception that the measurement clock and data recovery unit high-frequency corner bandwidth is 4 MHz.

138.3.2.2 Skew Constraints for 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8

Skew and Skew Variation are defined in 80.5 and 116.5 and specified at the points SP1 to SP6 shown in Figure 80–9 and Figure 116–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 80.5 and 116.5.

138.4 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that may provide control and status information for and about the PMD. If the MDIO interface is implemented, the mapping of MDIO control variables to PMD control variables shall be as shown in Table 138–4, and the mapping of MDIO status variables to PMD status variables shall be as shown in Table 138–5.

Table 138–4—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1 register	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable register	1.9.0	PMD_global_transmit_disable
PMD transmit disable $n-1$ to PMD transmit disable 0	PMD transmit disable register	1.9. n to 1.9.1	PMD_transmit_disable_ $n-1$ to PMD_transmit_disable_0

Table 138–5—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1 register	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2 register	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2 register	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect register	1.10.0	PMD_global_signal_detect
PMD receive signal detect $n-1$ to PMD receive signal detect 0	PMD receive signal detect register	1.10. n to 1.10.1	PMD_signal_detect_ $n-1$ to PMD_signal_detect_0

138.5 PMD functional specifications

The 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

138.5.1 PMD block diagram

The PMD block diagram for 200GBASE-SR8 is shown in Figure 138–2. The block diagrams for 200GBASE-SR4, 100GBASE-SR2, and 50GBASE-SR are equivalent to Figure 138–2, but for four lanes, two lanes, and one lane per direction, respectively.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a multimode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 138.8 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 138.10.3). Unless specified otherwise, all receiver measurements and tests defined in 138.8 are made at TP3.

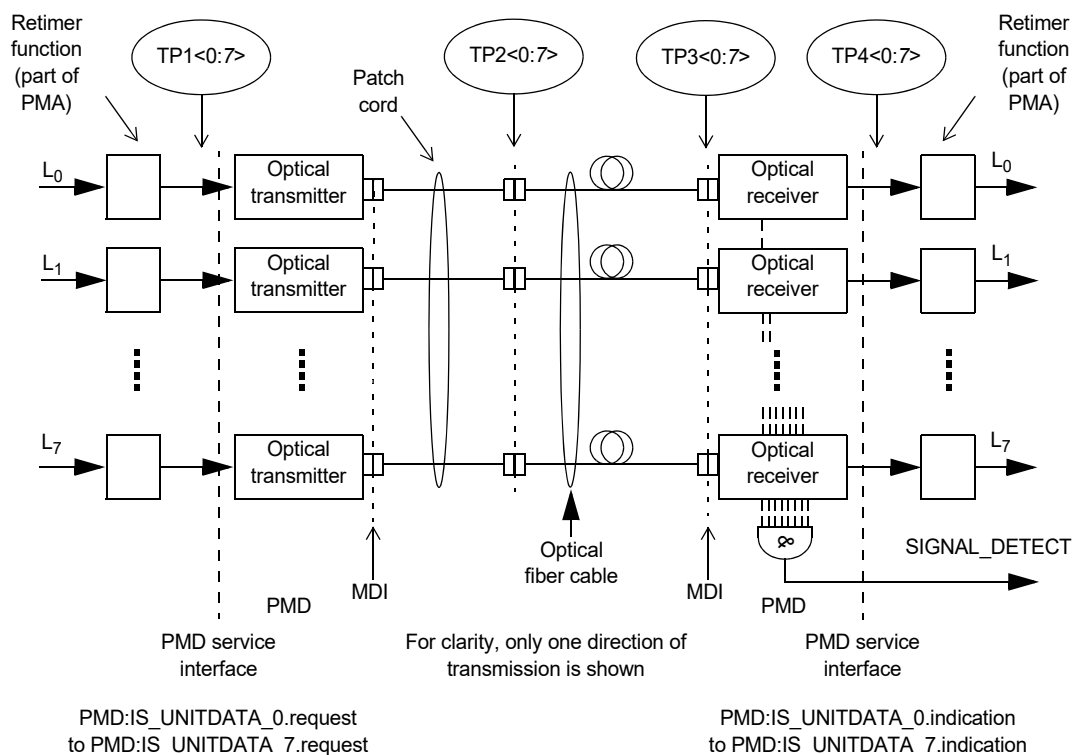


Figure 138–2—Block diagram for 400GBASE-SR8 transmit/receive paths

TP1<0:7> and TP4<0:7> are optional reference points that may be useful to implementers for testing components (these test points will not typically be accessible in an implemented system).

138.5.2 PMD transmit function

The PMD Transmit function shall convert the one, two, four, or eight symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_0.request into one, two, four, or eight separate optical signals. The 50GBASE-SR PMD has a single symbol stream, hence $i = 0$. The 100GBASE-SR2 PMD has two parallel symbol streams, hence $i = 0$ to 1. The 200GBASE-SR4 PMD has four parallel symbol streams, hence $i = 0$ to 3. The 400GBASE-SR8 PMD has eight parallel symbol streams, hence $i = 0$ to 7. Each optical signal shall then be delivered to the MDI, which contains one, two, four, or eight parallel light paths for transmit, according to the transmit optical specifications in this clause. The four optical power levels in the signal in order from lowest to highest shall correspond to tx_symbol values zero, one, two, and three, respectively.

Editor's note (to be removed after D3.1): According to the resolution of comment I-19, the phrase "signal streams requested by the PMD service interface" was to be changed to "signals requested by the PMD service interface". However, the editor's understanding is that the change should be to "symbol streams requested by the PMD service interface" instead.

138.5.3 PMD receive function

The PMD Receive function shall convert the one, two, four, or eight parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_7.indication, all according to the receive optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to rx_symbol values zero, one, two, and three, respectively.

138.5.4 PMD global signal detect function

The PMD global signal detect function shall report the state of SIGNAL_DETECT via the PMD service interface. The SIGNAL_DETECT parameter is signaled continuously, while the PMD:IS_SIGNAL.indication message is generated when a change in the value of SIGNAL_DETECT occurs. The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the inter-sublayer service interface primitives defined in 80.3.

SIGNAL_DETECT shall be a global indicator of the presence of optical signals on all lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 138–6. The PMD receiver is not required to verify whether a compliant 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Table 138–6—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ -30 dBm	FAIL
For all lanes; [(Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 138–8) AND (compliant 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 signal input)]	OK
All other conditions	Unspecified

As a consequence of the requirements for the setting of the SIGNAL_DETECT parameter, implementations need to provide adequate margin between the input optical power level at which the SIGNAL_DETECT parameter is set to OK, and the inherent noise level of the PMD including the effects of crosstalk, power supply noise, etc.

Various implementations of the Signal Detect function are permitted by this standard, including implementations that generate the SIGNAL_DETECT parameter values in response to the amplitude of the modulation of the optical signal and implementations that respond to the average optical power of the modulated optical signal.

138.5.5 PMD lane-by-lane signal detect function

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each PMD_signal_detect_*i*, where *i* represents the lane number in the range 0:7, shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 138–6.

138.5.6 PMD reset function

If the MDIO interface is implemented, and if PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

138.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

- a) When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 138–8.
- b) If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

138.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitter in each lane to be selectively disabled.

- a) When a PMD_transmit_disable_*i* variable (where *i* represents the lane number in the range 0:7) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 138–8.
- b) If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_*i* to one, turning off the optical transmitter in each lane.

If the optional PMD lane-by-lane transmit disable function is not implemented in MDIO, an alternative method may be provided to independently disable each transmit lane.

138.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

If the MDIO interface is implemented, PMD_fault shall be mapped to the fault bit as specified in 45.2.1.2.3.

138.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set PMD_transmit_fault to one.

If the MDIO interface is implemented, PMD_transmit_fault shall be mapped to the transmit fault bit as specified in 45.2.1.7.4

138.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

If the MDIO interface is implemented, PMD_receive_fault shall be mapped to the receive fault bit as specified in 45.2.1.7.5.

138.6 Lane assignments

There are no lane assignments (within a group of transmit or receive lanes) for 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8. While it is expected that a PMD will map electrical lane *i* to optical lane *i* and vice versa, there is no need to define the physical ordering of the lanes, as the RS-FEC or PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in 138.10.3.1.

138.7 PMD to MDI optical specifications for 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8

The operating range for the 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 PMDs is defined in Table 138–7. A compliant PMD operates on 50/125 µm multimode fibers, type A1a.2 (OM3), type A1a.3 (OM4), or type A1a.4 (OM5), according to the specifications defined in Table 138–14. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 50GBASE-SR PMD operating at 120 m meets the operating range requirement of 0.5 m to 100 m).

Table 138–7—Operating range

PMD type	Required operating range ^a
50GBASE-SR	0.5 m to 70 m for OM3
100GBASE-SR2	0.5 m to 100 m for OM4
200GBASE-SR4	
400GBASE-SR8	

^aThe PCS FEC correction function may not be bypassed for any operating distance.

138.7.1 Transmitter optical specifications

Each lane of a 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 transmitter shall meet the specifications in Table 138–8 per the definitions in 138.8.

Table 138–8—Transmit characteristics

Description	Value	Unit
Signaling rate, each lane (range)	26.5625 ± 100 ppm	GBd
Modulation format	PAM4	—
Center wavelength (range)	840 to 860	nm
RMS spectral width ^a (max)	0.6	nm
Average launch power, each lane (max)	4	dBm
Average launch power, each lane (min)	−6.5	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	3	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) ^b	−4.5	dBm
Launch power in OMA _{outer} minus TDECQ (min)	−5.9	dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	4.5	dB
TDECQ − $10\log_{10}(C_{eq})^c$, each lane (max)	4.5	dB
Average launch power of OFF transmitter, each lane (max)	−30	dBm
Extinction ratio, each lane (min)	3	dB
Transmitter transition time, each lane (max)	34	ps
RIN ₁₂ OMA (max)	−128	dB/Hz
Optical return loss tolerance (max)	12	dB
Encircled flux ^d	$\geq 86\%$ at 19 μm $\leq 30\%$ at 4.5 μm	—

^aRMS spectral width is the standard deviation of the spectrum.

^bEven if the TDECQ < 1.4 dB, the OMA (min) has to exceed this value.

^c C_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

^dIf measured into type A1a.2 or type A1a.3, or A1a.4, 50 μm fiber, in accordance with IEC 61280-1-4:2009.

138.7.2 Receiver optical specifications

Each lane of a 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 receiver shall meet the specifications in Table 138–9 per the definitions in 138.8.

Table 138–9—Receive characteristics

Description	Value	Unit
Signaling rate, each lane (range)	26.5625 ± 100 ppm	GBd
Modulation format	PAM4	—
Center wavelength (range)	840 to 860	nm

Table 138–9—Receive characteristics (*continued*)

Description	Value	Unit
Damage threshold ^a (min)	5	dBm
Average receive power, each lane (max)	4	dBm
Average receive power, each lane ^b (min)	−8.4	dBm
Receive power, each lane (OMA _{outer}) (max)	3	dBm
Receiver reflectance (max)	−12	dB
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	−3.4	dBm
Receiver sensitivity (OMA _{outer}), each lane ^d (max)	Equation (138–1)	dBm
Conditions of stressed receiver sensitivity test: ^e		
Stressed eye closure for PAM4 (SECQ), lane under test	4.5	dB
SECQ − 10log ₁₀ (C _{eq}) ^f (max), lane under test	4.5	dB
OMA _{outer} of each aggressor lane ^g	3	dBm

^aThe receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level on one lane. The receiver does not have to operate correctly at this input power.

^bAverage receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^cMeasured with conformance test signal at TP3 (see 138.8.10) for the BER specified in 138.1.1.

^dReceiver sensitivity is optional and is defined for a transmitter with a value of SECQ up to 4.5 dB.

^eThese test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^fC_{eq} is a coefficient defined in 121.8.5.3, which accounts for the reference equalizer noise enhancement.

^gOnly applies to 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8.

138.7.3 Illustrative link power budget

An illustrative power budget and penalties for 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 channels are shown in Table 138–10.

Table 138–10—Illustrative link power budget

Parameter	OM3	OM4	OM5	Unit
Effective modal bandwidth at 850 nm ^a	2000	4700		MHz.km
Power budget (for max TDECQ)		6.5		dB
Operating distance	0.5 to 70	0.5 to 100		m
Channel insertion loss ^b	1.8	1.9		dB
Allocation for penalties ^c (for max TDECQ)		4.6		dB
Additional insertion loss allowed	0.1	0		dB

^aPer IEC 60793-2-10.

^bThe channel insertion loss is calculated using the maximum distance specified in Table 138–7 and cabled optical fiber attenuation of 3.5 dB/km at 850 nm plus an allocation for connection and splice loss given in 138.10.2.2.1.

^cLink penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

138.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

138.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 138–12 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 138–12 may be used to perform that test. The test patterns used in this clause are shown in Table 138–11.

Table 138–11—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.4
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle	119.2.4.9
6	SSPRQ	120.5.11.2.3

Table 138–12—Test-pattern definitions and related subclauses

Parameter	Pattern	Related subclause
Wavelength, spectral width	3, 4, 5, 6, or valid 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 signal	138.8.2
Average optical power	3, 4, 5, 6, or valid 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 signal	138.8.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	138.8.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	138.8.5
Extinction ratio	4 or 6	138.8.6
Transmitter transition time	Square wave or 6	138.8.7
RIN ₁₂ OMA	Square wave	138.8.8
Stressed receiver sensitivity	3, 5, or valid 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 signal	138.8.10
Stressed eye closure (SEC), calibration	6	138.8.10

138.8.1.1 Multi-lane testing considerations

Stressed receiver sensitivity is defined for each lane at the BER specified in 138.1.1. Measurements with Pattern 3 (PRBS31Q) allow lane-by-lane BER measurements. In stressed receiver sensitivity measurements, unstressed lanes may be created by setting the power at the receiver under test well above its sensitivity and/or not stressing those lanes with ISI and jitter, or by other means. Each receive lane is stressed in turn while all are operated. All aggressor lanes are operated as specified. Where relevant, parameters are defined with all co-propagating and counter-propagating lanes operational so that crosstalk effects are included. Where not otherwise specified, the maximum amplitude ($\text{OMA}_{\text{outer}}$) for a particular situation is used. Alternative test methods that generate equivalent results may be used. While the lanes in a particular direction may share a common clock, the Tx and Rx directions are not synchronous to each other. If Pattern 3 is used for the lanes not under test using a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane so that the symbols on each lane are not correlated within the PMD.

138.8.2 Center wavelength and spectral width

The center wavelength and RMS spectral width of each optical lane shall be within the range given in Table 138–8 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns specified in Table 138–12.

138.8.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 138–8 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns defined in Table 138–12.

138.8.4 Outer Optical Modulation Amplitude ($\text{OMA}_{\text{outer}}$)

The $\text{OMA}_{\text{outer}}$ of each lane shall be within the limits given in Table 138–8 if measured as defined in 121.8.4.

138.8.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

TDECQ is a measure of each optical transmitter's vertical eye closure as measured through an optical to electrical converter (O/E) with a bandwidth equivalent to a combined reference receiver and worst case optical channel, and equalized with the reference equalizer specified in 138.8.5.1. Table 138–9 specifies the test pattern to be used for measurement of TDECQ.

The TDECQ and $\text{TDECQ} - 10\log_{10}(C_{\text{eq}})$ of each lane shall be within the limits given in Table 138–8 if measured using the methods specified in 121.8.5, with the following exceptions:

- The polarization rotator and test fiber shown in Figure 121–4 are not used.
- The optical channel requirements in 121.8.5.2 do not apply. For 400GBASE-SR8, the optical splitter and variable reflector are adjusted so that each transmitter is tested with an optical return loss equal to the “Optical return loss tolerance (max)” given in Table 138–8.
- The combination of the O/E converter and the oscilloscope used to measure the optical waveform has a 3 dB bandwidth of approximately 11.2 GHz with a fourth-order Bessel-Thomson response to at least 1.5×22.4 GHz and at frequencies above 1.5×22.4 GHz the response should not exceed –24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The reference equalizer to be used for TDECQ for 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 is specified in 138.8.5.1.

138.8.5.1 TDECQ reference equalizer

The reference equalizer for 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 is a 5-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period. A functional model of the reference equalizer is shown in Figure 138–3. The sum of the equalizer tap coefficients is equal to 1. Tap 1, tap 2, or tap 3, has the largest magnitude tap coefficient, which is constrained to be at least 0.8.

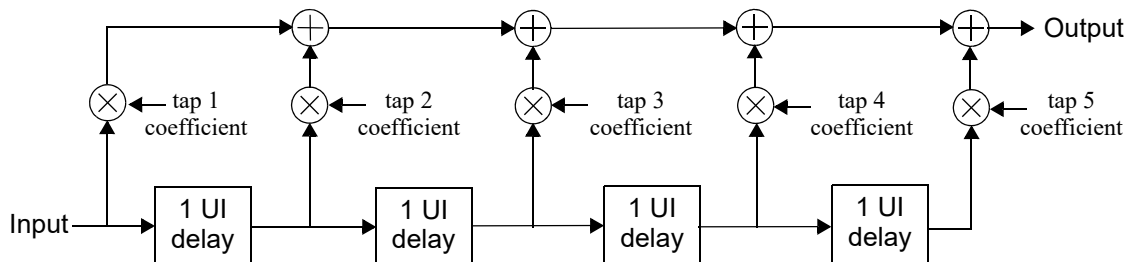


Figure 138–3—TDECQ reference equalizer functional model

NOTE—This reference equalizer is part of the TDECQ test and does not imply any particular receiver implementation.

138.8.6 Extinction ratio

The extinction ratio of each lane shall be within the limits given in Table 138–8 if measured using the methods specified in 121.8.6.

138.8.7 Transmitter transition time

The transmitter transition time of each lane shall be within the limits given in Table 138–8 if measured using a test pattern specified for transmitter transition time in Table 138–12.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} , or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively, as measured through an O/E converter and oscilloscope with a combined 3 dB bandwidth of approximately 13.28125 GHz with a fourth-order Bessel-Thomson response to at least 1.5×26.5625 GHz and at frequencies above 1.5×26.5625 GHz the response should not exceed –24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 138.8.4), with the exception that the square wave test pattern can be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

138.8.8 Relative intensity noise (RIN_{12OMA})

RIN shall be as defined by the measurement methodology of 52.9.6 with the following exceptions:

- The optical return loss is 12 dB.
- The upper –3 dB limit of the measurement apparatus is to be approximately equal to the signaling rate (i.e., 26.6 GHz).
- The test pattern is according to Table 138–12.

138.8.9 Receiver sensitivity

Receiver sensitivity is optional and is defined for a transmitter with a value of SECQ up to 4.5 dB. Receiver sensitivity should meet Equation (138–1), which is illustrated in Figure 138–4.

$$RS = \max(-6.5, SECQ - 7.9) \quad (\text{dBm}) \quad (138-1)$$

where

RS is the receiver sensitivity

$SECQ$ is the SECQ of the transmitter used to measure the receiver sensitivity

The normative requirement for receivers is stressed receiver sensitivity.

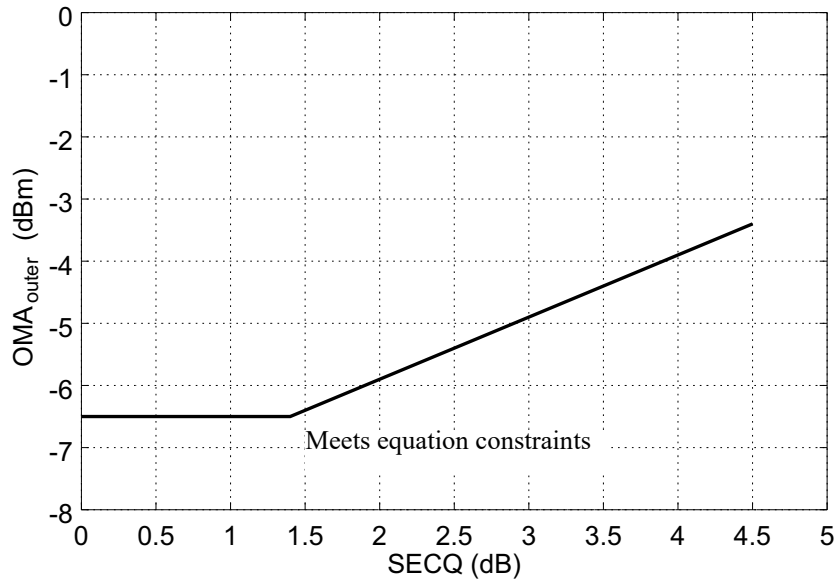


Figure 138–4—Illustration of receiver sensitivity

138.8.10 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 138–9 if measured using the methodology defined in 121.8.10.1 and 121.8.10.3, with the conformance test signal at TP3 as described in 121.8.5.2, with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 138.8.5, except that the combination of the O/E converter and the oscilloscope has a 3 dB bandwidth of approximately 13.28125 GHz with a fourth-order Bessel-Thomson response to at least 1.5×26.5625 GHz and at frequencies above 1.5×26.5625 GHz the response should not exceed –24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response. The optical splitter and variable reflector shown in Figure 121–4 are not used, and the transition time is no greater than the value specified in Table 138–8.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the RIN_{12OMA} of the SRS test source should be no greater than the $RIN_{12OMA}(\text{max})$ specified for the transmit characteristics in Table 138–8.
- The test patterns used for stressed receiver sensitivity are specified in Table 138–12.
- The signaling rate, the required stressed eye closure (SECQ), and the maximum $SECQ - 10\log_{10}(C_{eq})$ of the stressed receiver conformance test signal is as specified in Table 138–9.

- The applied sinusoidal jitter is specified in 138.8.10.1.
- For 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8, the OMA_{outer} of the aggressor lanes is specified in Table 138–9.

The BER is required to be met for each lane under test on its own. Stressed receiver sensitivity is defined with all transmit and receive lanes in operation. Any of the patterns specified for stressed receiver sensitivity in Table 138–12 is sent from the transmit section of the PMD under test. The signal being transmitted is asynchronous to the received signal.

138.8.10.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 138–13.

Table 138–13—Applied sinusoidal jitter

Frequency range	Sinusoidal jitter peak-to-peak (UI)
$f < 40 \text{ kHz}$	Not specified
$40 \text{ kHz} < f \leq 4 \text{ MHz}$	$2 \times 10^5 \text{ Hz}/f$
$4 \text{ MHz} < f \leq 10 \text{ LB}^a$	0.05

^a LB = loop bandwidth; upper frequency bound for added sinusoidal jitter should be at least 10 times the loop bandwidth of the receiver being tested.

138.9 Safety, installation, environment, and labeling

138.9.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

138.9.2 Laser safety

50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8 optical transceivers shall conform to Hazard Level 1M laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.²²⁹

²²⁹A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

138.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

138.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate, in the literature associated with the PHY, the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

138.9.5 Electromagnetic emission

A system integrating a 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

138.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

138.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8).

Labeling requirements for Hazard Level 1M lasers are given in the laser safety standards referenced in 138.9.2.

138.10 Fiber optic cabling model

The fiber optic cabling (channel) contains 1, 2, 4, or 8 optical fibers for each direction to support 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8, respectively. The fiber optic cabling interconnects the transmitters at the MDI on one end of the channel to the receivers at the MDI on the other end of the channel. As defined in 138.10.3, the optical lanes appear in defined locations at the MDI but the locations are not assigned specific lane numbers within this standard because any transmitter lane may be connected to any receiver lane.

138.10.1 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 138–5.

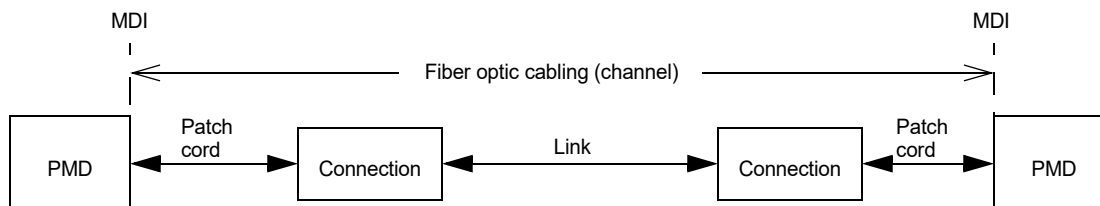


Figure 138–5—Fiber optic cabling model

The channel insertion loss is given in Table 138–14. A channel may contain additional connectors as long as the optical characteristics of the channel (such as attenuation, modal dispersion, reflections and losses of all connectors and splices) meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-1 one-cord method. As OM4 and OM5 optical fiber meet the requirements for OM3, a channel compliant to the “OM3” column may use OM4 or OM5 optical fiber, or a combination of OM3, OM4, and OM5. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term channel is used here for consistency with generic cabling standards.

Table 138–14—Fiber optic cabling (channel) characteristics

Description	OM3	OM4	OM5	Unit
Operating distance (max)	70	100		m
Cabling Skew (max) ^a	79			ns
Cabling Skew Variation ^{ab} (max)	2.4			ns
Channel insertion loss ^c (max)	1.8	1.9		dB
Channel insertion loss (min)	0			dB

^aApplies only to 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8.

^bAn additional 400 ps of Skew Variation could be caused by wavelength changes, which are attributable to the transmitter not the channel.

^cThese channel insertion loss values include cable loss plus 1.5 dB allocated for connection and splice loss, over the wavelength range 840 nm to 860 nm.

138.10.2 Characteristics of the fiber optic cabling (channel)

The fiber optic cabling shall meet the specifications defined in Table 138–14. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

138.10.2.1 Optical fiber cable

The fiber contained within the fiber optic cabling shall comply with the specifications and parameters of Table 138–15. A variety of multimode cable types may satisfy these requirements, provided the resulting channel also meets the specifications of Table 138–14.

Table 138–15—Optical fiber and cable characteristics

Description	OM3 ^a	OM4 ^b	OM5 ^c	Unit
Nominal core diameter	50			μm
Nominal fiber specification wavelength	850			nm
Effective modal bandwidth (min) ^d	2000	4700		MHz.km
Cabled optical fiber attenuation (max)	3.5			dB/km
Zero dispersion wavelength (λ_0)	$1295 \leq \lambda_0 \leq 1340$		$1297 \leq \lambda_0 \leq 1328$	nm
Chromatic dispersion slope (max) (S_0)	0.105 for $1295 \leq \lambda_0 \leq 1310$ and $0.000375 \times (1590 - \lambda_0)$ for $1310 \leq \lambda_0 \leq 1340$		$-412/(840(1 - (\lambda_0/840)^4))$	ps/nm ² km

^aIEC 60793-2-10 type A1a.2

^bIEC 60793-2-10 type A1a.3

^cIEC 60793-2-10 type A1a.4

^dWhen measured with the launch conditions specified in Table 138–8

138.10.2.2 Optical fiber connection

An optical fiber connection, as shown in Figure 138–5, consists of a mated pair of optical connectors.

138.10.2.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 1.5 dB total connection and splice loss. For example, this allocation supports three connections with an average insertion loss per connection of 0.5 dB. Connections with lower loss characteristics may be used provided the requirements of Table 138–14 are met. However, the loss of a single connection shall not exceed 0.75 dB.

138.10.2.2.2 Maximum discrete reflectance

The maximum discrete reflectance shall be less than –20 dB.

138.10.3 Medium Dependent Interface (MDI)

The 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, or 400GBASE-SR8 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 138–5). Examples of an MDI include the following:

- a) PMD with a connectorized fiber pigtail plugged into an adapter;
- b) PMD receptacle.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 138.5.1, not at the MDI.

138.10.3.1 Optical lane assignments for 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8

The two transmit and two receive optical lanes of 100GBASE-SR2 shall occupy the positions depicted in Figure 138–6 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains four active lanes within 12 total positions. The eight center positions are unused. The transmit optical lanes occupy the leftmost two positions. The receive optical lanes occupy the rightmost two positions.

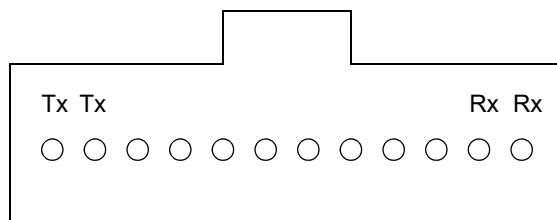


Figure 138–6—Optical lane assignments for 100GBASE-SR2

The four transmit and four receive optical lanes of 200GBASE-SR4 shall occupy the positions depicted in Figure 138–7 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains eight active lanes within 12 total positions. The four center positions are unused. The transmit optical lanes occupy the leftmost four positions. The receive optical lanes occupy the rightmost four positions.

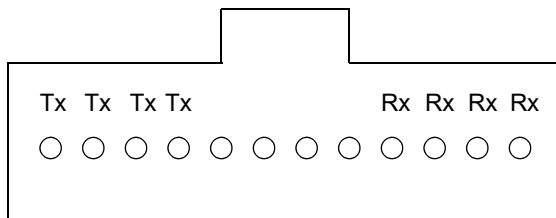


Figure 138–7—Optical lane assignments for 200GBASE-SR4

400GBASE-SR8 has two optical lane assignment options: Option A for a dual-row twelve-fiber interface, and option B for a single-row sixteen-fiber interface.

For a dual-row twelve-fiber interface, the eight transmit and eight receive optical lanes of 400GBASE-SR8 shall occupy the positions depicted in Figure 138–8 (option A) when looking into the MDI receptacle with the connector keyway feature on top. The interface contains sixteen active lanes within 24 total positions.

The four center positions in each row are unused. The transmit optical lanes occupy the leftmost four positions in each row. The receive optical lanes occupy the rightmost four positions in each row.

For a single-row sixteen-fiber interface, the eight transmit and eight receive optical lanes of 400GBASE-SR8 shall occupy the positions depicted in Figure 138–8 (option B) when looking into the MDI receptacle with the connector keyway feature on top. The interface contains sixteen active lanes within sixteen total positions. The transmit optical lanes occupy the leftmost eight positions. The receive optical lanes occupy the rightmost eight positions.

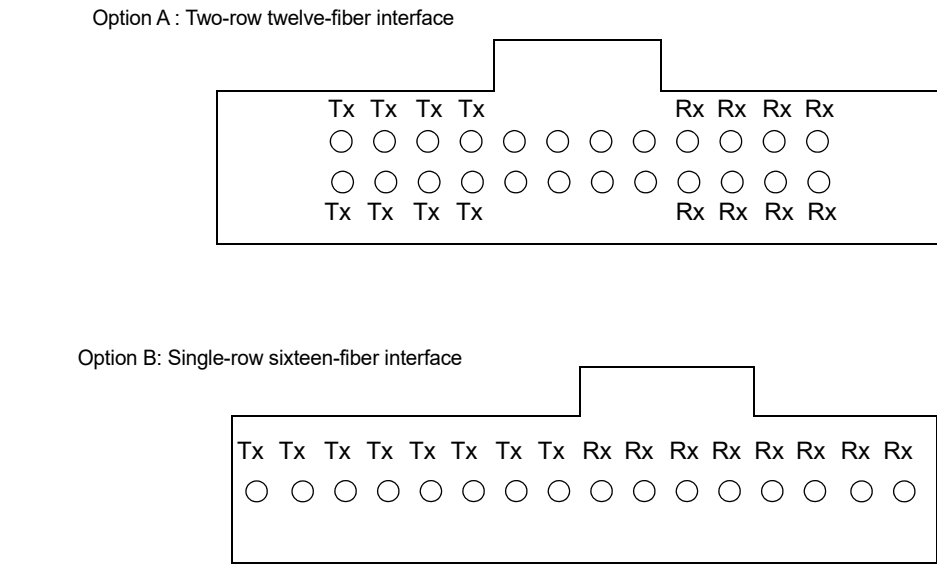


Figure 138–8—Optical lane assignments for 400GBASE-SR8

138.10.3.2 MDI requirements for 50GBASE-SR

The MDI shall optically mate with the compatible plug on the optical fiber cabling. For 50GBASE-SR, when the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-022-2 for performance grade Bm/2m.

138.10.3.3 MDI requirements for 100GBASE-SR2 and 200GBASE-SR4

The MDI shall optically mate with the compatible plug on the optical fiber cabling. For 100GBASE-SR2 and 200GBASE-SR4 the MDI adapter or receptacle shall meet the dimensional specifications for interface 7-1-3: *MPO adapter interface - opposed keyway configuration*, or interface 7-1-10: *MPO active device receptacle, flat interface*, as defined in IEC 61754-7-1. The plug terminating the optical fiber cabling shall meet the dimensional specifications of interface 7-1-4: *MPO female plug connector, flat interface for 2 to 12 fibers*, as defined in IEC 61754-7-1. Figure 138–9 shows an MPO female plug connector with flat interface, and an MDI. The MDI connection shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-022-2 for performance grade Bm/2m.

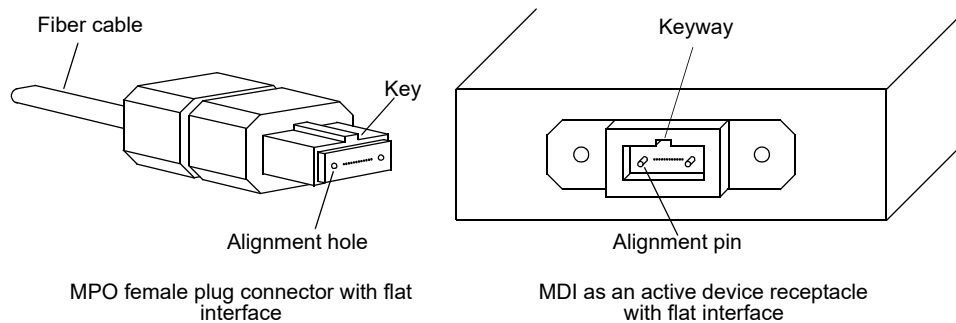


Figure 138–9—MPO female plug with flat interface and MDI active device receptacle with flat interface

138.10.3.4 MDI requirements for 400GBASE-SR8

The MDI shall optically mate with the compatible plug on the optical fiber cabling. The MDI connection shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-022-2 for performance grade Bm/2m. 400GBASE-SR8 has two optical lane assignment options: Option A for a dual-row twelve-fiber interface, and option B for a single-row sixteen-fiber interface.

For Option A, the MDI adapter or receptacle shall meet the dimensional specifications for interface 7-2-3: *MPO adapter interface - opposed keyway configuration*, or interface 7-2-10: *MPO active device receptacle, flat interface*, as defined in IEC 61754-7-2. The plug terminating the optical fiber cabling shall meet the dimensional specifications of interface 7-2-4: *MPO female plug connector, flat interface for 16 to 24 fibers*, as defined in IEC 61754-7-2. The MPO female plug connector and MDI are structurally similar to those depicted in Figure 138–9, but with two rows of fibers.

For Option B, the MDI adapter or receptacle shall meet the dimensional specifications for designation FOCIS 18 A-1-0, or designation FOCIS 18 R-1x16-1-0-1-2-0, as defined in ANSI/TIA-604-18-A. The plug terminating the optical fiber cabling shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-0-2-2-0, as defined in ANSI/TIA-604-18-A. The MPO female plug connector and MDI are structurally similar to those depicted in Figure 138–9, but with sixteen fibers and an offset keyway as well as a different pin diameter and location.

138.11 Protocol implementation conformance statement (PICS) proforma for Clause 138, Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8²³⁰

138.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 138, Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

138.11.2 Identification

138.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

138.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 138, Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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²³⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

138.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*SR	50GBASE-SR PMD	138.7	Device supports requirements for 50GBASE-SR PHY	O	Yes [] No []
*SR2	100GBASE-SR2 PMD	138.7	Device supports requirements for 100GBASE-SR2 PHY	O	Yes [] No []
*SR4	200GBASE-SR4 PMD	138.7	Device supports requirements for 200GBASE-SR4 PHY	O	Yes [] No []
*SR8	400GBASE-SR8 PMD	138.7	Device supports requirements for 400GBASE-SR8 PHY	O	Yes [] No []
*SR8A	400GBASE-SR8 MDI option A	138.10.3.4	Device has MDI with dual-row twelve-fiber interface	SR8:O.1	Yes [] No []
*SR8B	400GBASE-SR8 MDI option B	138.10.3.4	Device has MDI with single-row sixteen-fiber interface	SR8:O.1	Yes [] No []
*INS	Installation / cable	138.10.1	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	138.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	138.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	138.3.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	138.3.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	138.4	Registers and interface supported	O	Yes [] No []

138.11.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8

138.11.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 50GBASE-R, 100GBASE-R, 200GBASE-R, or 400GBASE-R PCS and PMA	138.1		M	Yes []
F2	Integration with management functions	138.1		O	Yes [] No []
F3	Bit error ratio	138.1.1	Meets the BER specified in 138.1.1	M	Yes []
F4	Transmit function	138.5.2	Conveys bits from PMD service interface to MDI	M	Yes []
F5	Mapping between optical signal and logical signal for transmitter	138.5.2	Optical power levels from lowest to highest correspond to tx_symbol values zero, one, two, and three, respectively	M	Yes []
F6	Receive function	138.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F7	Conversion of n optical signals to n electrical signals	138.5.3	For delivery to the PMD service interface	M	Yes []
F8	Mapping between optical signal and logical signal for receiver	138.5.3	Optical power levels from lowest to highest correspond to rx_symbol values zero, one, two, and three, respectively	M	Yes []
F9	Global Signal Detect function	138.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication(SIGNAL_DETECT)	M	Yes []
F10	Global Signal Detect behavior	138.5.4	SIGNAL_DETECT is a global indicator of the presence of optical signals on all n lanes	M	Yes []
F11	Lane-by-lane Signal Detect function	138.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 138–6	MD:O	Yes [] No [] N/A []
F12	PMD reset function	138.5.6	Resets the PMD sublayer	MD:O	Yes [] No [] N/A []

138.11.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	138.4		MD:M	Yes [] N/A []
M2	Global transmit disable function	138.5.7	Disables all of the optical transmitters with the PMD_global_transmit_disable variable	MD:O	Yes [] No [] N/A []
M3	PMD_lane_by_lane_transmit_disable function	138.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	MD:O	Yes [] No [] N/A []
M4	PMD lane-by-lane transmit disable	138.5.8	Disables each optical transmitter independently if M3 = No	!MD:O	Yes [] No [] N/A []
M5	PMD_fault function	138.5.9	Sets PMD_fault to one if any local fault is detected	MD:O	Yes [] No [] N/A []
M6	PMD_transmit_fault function	138.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	MD:O	Yes [] No [] N/A []
M7	PMD_receive_fault function	138.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	MD:O	Yes [] No [] N/A []

138.11.4.3 PMD to MDI optical specifications

Item	Feature	Subclause	Value/Comment	Status	Support
S1	Transmitter meets specifications in Table 138–8	138.7.1	Per definitions in 138.8	M	Yes []
S2	Receiver meets specifications in Table 138–9	138.7.2	Per definitions in 138.8	M	Yes []

138.11.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	138.8	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and spectral width	138.8.2	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	138.8.3	Per IEC 61280-1-1	M	Yes []
OM4	OMA _{outer} measurements	138.8.4	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	138.8.5	Each lane	M	Yes []
OM6	Extinction ratio	138.8.6	Per IEC 61280-2-2	M	Yes []
OM7	Transmitter transition time	138.8.7	Each lane	M	Yes []
OM8	RIN ₁₂ OMA measurement procedure	138.8.8	Each lane	M	Yes []
OM9	Stressed receiver sensitivity	138.8.10	See 138.8.10	M	Yes []

138.11.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	138.9.1	Conforms to J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1M	138.9.2	Conforms to Hazard Level 1M laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	138.9.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

138.11.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	138.10	Meets requirements specified in Table 138–14	INS:M	Yes [] N/A []
OC2	Optical fiber characteristics	138.10.1	Per Table 138–15	INS:M	Yes [] N/A []
OC3	Maximum discrete reflectance	138.10.2.2	Less than –20 dB	INS:M	Yes [] N/A []
OC4	MDI layout for 100GBASE-SR2	138.10.3.1	Optical lane assignments per Figure 138–6	SR2:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
OC5	MDI layout for 200GBASE-SR4	138.10.3.1	Optical lane assignments per Figure 138–7	SR4:M	Yes [] N/A []
OC6	MDI layout for 400GBASE-SR8 option A	138.10.3.1	Optical lane assignments per Figure 138–8	SR8A:M	Yes [] N/A []
OC7	MDI layout for 400GBASE-SR8 option B	138.10.3.1	Optical lane assignments per Figure 138–8	SR8B:M	Yes [] N/A []
OC8	MDI mating, 50GBASE-SR	138.10.3.2	MDI optically mates with plug on the cabling, performance grade Bm/2m	SR:M	Yes [] N/A []
OC9	MDI requirements for 50GBASE-SR	138.10.3.2	Per IEC 61753-1 and IEC 61753-022-2	INS:M	Yes [] N/A []
OC10	MDI mating, 100GBASE-SR2 and 200GBASE-SR4	138.10.3.3	MDI optically mates with plug on the cabling	(SR2 or SR4):M	Yes [] N/A []
OC11	MDI mating 400GBASE-SR8	138.10.3.4	MDI optically mates with plug on the cabling	SR8:M	Yes [] N/A []
OC12	MDI dimensions for 100GBASE-SR2 and 200GBASE-SR4	138.10.3.3	Per IEC 61754-7-1 interface 7-1-3 or interface 7-1-10	(SR2 or SR4):M	Yes [] N/A []
OC13	Cabling connector dimensions for 100GBASE-SR2 and 200GBASE-SR4	138.10.3.3	Per IEC 61754-7-1 interface 7-1-4	INS*(SR2 or SR4):M	Yes [] N/A []
OC14	MDI dimensions for 400GBASE-SR8	138.10.3.4	Per IEC 61754-7-2 interfaces 7-2-3 or 7-2-10, or per ANSI/TIA-604-18-A designation FOCIS 18 A-1-0 or FOCIS 18 R-1x16-1-0-1-2-0	SR8:M	Yes [] N/A []
OC15	Cabling connector dimensions for 400GBASE-SR8	138.10.3.4	Per IEC 61754-7-2 interface 7-2-4 or ANSI/TIA-604-18-A designation FOCIS 18 P-1x16-1-0-2-2-0	INS*SR8:M	Yes [] N/A []
OC16	MDI requirements for 100GBASE-SR2 and 200GBASE-SR4	138.10.3.3	Per IEC 61753-1 and IEC 61753-022-2, performance grade Bm/2m	INS*(SR2 or SR4):M	Yes [] N/A []
OC17	MDI requirements for 400GBASE-SR8	138.10.3.4	Per IEC 61753-1 and IEC 61753-022-2, performance grade Bm/2m	INS*SR8:M	Yes [] N/A []